

duction process ensures quality of end products. Monitoring therefore starts from design stage to assembly to soldering to printing. Advanced inline inspection systems, especially high-speed and high-resolution AOI systems, are the norm today. AOI system is used for inspecting the PCBs after the assembly is done. State of the art AOI systems include capabilities to inspect 01005-size components, high-speed pre- and post-reflow inspection capabilities, advanced vision technologies, ability to inspect any component or feature such as chips, integrated circuits, empty locations, lead-free solder joints, clips and screws. Leaptech Corporation's Omron (Japan)-make AOI machine VT-RNS-II is an ideal inspection system for PCB assembly. It can accurately inspect all SMD components, solder quality, etc, in any circumstances—after solder paste printing, after component placement, or after reflow soldering—and hence is suitable for any sector application. Its price starts from \$45,000 for a tabletop unit.

Green is the colour

As 'green' manufacturing gains currency, customers are looking for environment-friendly and energy-efficient SMT equipment, which also meet with the requirements of directives such as restriction of hazardous substances directive and waste electrical and electronic equipment directive. As a result, a variety of equipment are suitable for lead-free production. For example, SMT filters that survive higher solder reflow temperature profiles to make PCBs RoHS compliant are considered a green machine. OK International's SmartHeat conduction products satisfy these needs with their inherent capability to solder faster at safer temperatures.

When will India manufacture?

Despite the demand for SMT equipment growing over the years, no

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domestic players manufacture such equipment, barring a handful of players who make wave soldering and reflow machines. But with the new manufacturing policy coming up, is there any possibility of SMT machines and equipment being manufactured in India? A variety of factors such as fiscal incentives being assured by the India government, competitive labour costs and hi-tech manufacturing knowledge are attracting investments and making India an ideal electronic manufacturing destination. Yet producing SMT machines in India is a far cry.

"At least in the next five years

manufacturing of SMT machines in India will not happen, as neither there is that kind of demand nor an environment has been created for manufacturers to take the risk," says Pradeep Tandon. "Manufacturing cannot happen because the supply chain does not exist in India, and the local demand is not yet high as compared to other countries. May be reflow ovens and printers shall be made in India, but pick-and-place shall not be made in India in the near future," opines Alok Gupta, director, Prosem Technology.

S.K. Jain, managing director, Sumitron Exports says, "Low-end technology machines like reflow ovens and soldering/desoldering stations are already being manufactured in India by some indigenous companies. However, these Indian equipment still do not match up with the machines being offered by foreign manufacturers. Also, we don't foresee the production of high-end machines like pick-and-place, AOI systems or X-ray inspection systems in India in the near future, even if the new manufacturing policy is in place. We need to put in huge amount of funds in R&D so as to be able to achieve that goal."

The cost factor

With India being a price sensitive country, the cost factor of SMT machines has been the most important hindrance in its market growth. How can these machines become cost-efficient? According to Prasad Gurjal, volume and exports will certainly help to bring down cost. Dilip Rane suggests that by practical operational benefits such as easy maintenance and high efficiency of the machines, along with important service elements such as customer support and product accessibility, customers can boost their productivity and save money.

Pradeep Tandon, however, believes that, though the latest SMT solutions are very costly, there is no doubt that the cost of new machines